Lecture 6 Notes:
Aristotle's *Posterior Analytics*, Book I

Central Theme

Aristotle's *Posterior Analytics*, Book I is the transition from valid inference to scientific knowledge.

In the *Prior Analytics*, Aristotle asked:

When do propositions force a conclusion?

In the *Posterior Analytics*, he asks:

When does a valid conclusion count as knowledge?

$$\boxed{\textit{Posterior Analytics}, \text{ Book I} = \text{Aristotle's theory of scientific demonstration}}$$

A syllogism gives a conclusion. A demonstration gives knowledge of the reason why.

1. From Syllogism to Demonstration

The *Prior Analytics* studied syllogism in general. The *Posterior Analytics* studies a special kind of syllogism: demonstration.

$$\textit{Prior Analytics} = \text{valid syllogism}$$

$$\textit{Posterior Analytics} = \text{scientific demonstration}$$

A demonstration is a syllogism productive of scientific knowledge.

$$\boxed{\text{Demonstration} = \text{a syllogism that produces scientific knowledge}}$$

Teaching Point

Not every valid syllogism is a demonstration.

A syllogism may be formally valid and still fail to produce scientific knowledge if its premises are not true, explanatory, primary, necessary, and appropriate.

$$\boxed{\text{Validity} \neq \text{demonstration}}$$

2. All Teaching Begins from Pre-Existent Knowledge

Aristotle begins with the claim that all instruction given or received by argument proceeds from pre-existent knowledge.

This applies to:

- mathematics,
- speculative sciences,
- syllogistic reasoning,
- induction,
- rhetoric,
- example,
- enthymeme.

Two Kinds of Pre-Existent Knowledge

Sometimes the learner must already know that something is.

$$\text{knowledge that something exists}$$

Sometimes the learner must already know what a term means.

$$\text{knowledge of meaning}$$

Sometimes both are required.

Examples

With triangle, one may need to understand what the word means.

$$\text{triangle} = \text{a three-sided rectilinear figure}$$

With unit, one must understand the meaning and also assume that such a thing exists.

$$\text{unit} = \text{meaning} + \text{existence}$$

Teaching Point

Learning never begins from absolutely nothing.

$\boxed{\text{Before learning, we must already know something.}}$

3. Aristotle and the Puzzle of the *Meno*

Aristotle's opening claim helps answer the problem made famous in Plato's *Meno*.

The puzzle is:

> If we already know, we do not need to learn. If we do not know, we cannot know what to look for.

Aristotle answers by distinguishing different senses of knowing.

A person may know something universally without yet recognizing its application to a particular case.

Example

A student may know:

Every triangle has angles equal to two right angles.

But the student may not yet know that this figure here is a triangle.

Once the student recognizes the particular as falling under the universal, the student comes to know the particular case.

Teaching Point

Learning moves from prior universal grasp to particular recognition.

$\boxed{\text{One may know universally and still learn the particular application.}}$

4. What Is Scientific Knowledge?

Aristotle says that we possess unqualified scientific knowledge when we know:

1. the cause on which the fact depends,
2. that it is the cause of that fact,
3. that the fact could not be otherwise.

$$\boxed{\text{Scientific knowledge} = \text{knowledge of the cause as cause}}$$

Scientific knowledge is not merely true belief. It is knowledge of why the truth must be so.

$$\boxed{\text{Scientific knowledge concerns what cannot be otherwise.}}$$

Teaching Point

To know scientifically is not simply to know that something is true. It is to know why it is necessarily true.

$$\text{knowledge that} \quad \neq \quad \text{knowledge why}$$

5. Demonstration as Scientific Syllogism

Demonstration is the kind of syllogism that produces scientific knowledge.

$$\boxed{\text{Demonstration} = \text{syllogism productive of scientific knowledge}}$$

The premises of a demonstration must be:

1. true
2. primary
3. immediate
4. better known than the conclusion
5. prior to the conclusion
6. causes of the conclusion

Teaching Point

A demonstration is not just a valid argument. It is a valid argument whose premises explain the conclusion.

$$\boxed{\text{Syllogism} + \text{explanatory premises} = \text{demonstration}}$$

6. Example: Validity Without Demonstration

Consider the following argument:

All stones are animals.

All humans are stones.

$\therefore$ All humans are animals.

The conclusion is true, and the form resembles a valid syllogism. But this is not a demonstration.

Why not?

- The premises are false.
- The premises are not explanatory.
- The middle term does not give the cause.
- The argument does not show why humans are animals.

Teaching Point

Demonstration requires more than a conclusion that happens to be true.

A true conclusion from bad premises is not scientific knowledge.

7. Prior and Better Known: To Us and By Nature

Aristotle distinguishes two senses of being prior and better known.

prior and better known to us

prior and better known by nature

Things closer to sense-perception are usually better known to us.

particulars, examples, visible cases

Causes and universals are prior and better known by nature.

principles, causes, universals

Whiteboard

To us	sense-near particulars
By nature	causes and universals

Teaching Point

Learning starts from what is clearer to us and aims at what is clearer by nature.

Education moves from familiar facts to explanatory causes.

8. Basic Truths

A basic truth is an immediate proposition.

$$\boxed{\text{Basic truth} = \text{immediate proposition}}$$

An immediate proposition has no prior proposition through which it is proved.

Basic truths are necessary because demonstration cannot go on forever. If every premise required another premise before it, no demonstration could ever be completed.

Teaching Point

Demonstration rests on starting points that are not themselves demonstrated.

$$\boxed{\text{Science requires indemonstrable first principles.}}$$

9. Thesis, Axiom, Hypothesis, and Definition

Aristotle distinguishes several kinds of starting points.

Thesis	an immediate basic truth not proved by the teacher
Axiom	a truth the learner must know to learn anything
Hypothesis	an assumption of existence or non-existence
Definition	a statement of what something means

Hypothesis and Definition

A hypothesis asserts that something is or is not.

There are units.

A definition states what something is without asserting its existence.

A unit is quantitatively indivisible.

Teaching Point

A definition lays down meaning; a hypothesis asserts existence.

$$\boxed{\text{To define what something is is not yet to assert that it exists.}}$$

10. Against Infinite Regress

Aristotle rejects the view that all knowledge requires demonstration.

If every premise had to be demonstrated by a prior premise, then knowledge would require an infinite regress.

$$P_1 \leftarrow P_2 \leftarrow P_3 \leftarrow P_4 \leftarrow \cdots$$

But an infinite regress cannot be traversed.

$$\boxed{\text{Infinite regress} \Rightarrow \text{no completed scientific knowledge}}$$

Therefore, not all knowledge is demonstrative. Some knowledge must be knowledge of immediate first principles.

Teaching Point

Demonstration is possible only because some truths are known without being demonstrated.

$$\boxed{\text{Not all knowledge is demonstrative.}}$$

11. Against Circular Demonstration

Aristotle also rejects circular demonstration.

Circular demonstration attempts to prove premises by means of the conclusion they are supposed to prove.

$$A \Rightarrow B$$

$$B \Rightarrow A$$

This destroys explanatory priority.

Why Circular Demonstration Fails

Demonstration must proceed from what is prior and better known to what is posterior and less known.

But in circular demonstration, the same thing becomes both prior and posterior.

$$\boxed{\text{The same thing cannot be both prior and posterior in the same demonstration.}}$$

Teaching Point

Circular demonstration may show reciprocal implication, but it does not give scientific explanation.

$$\boxed{\text{Science requires first principles, not explanatory circles.}}$$

12. Types of Attribute

Aristotle next asks what kind of attributes can be demonstrated scientifically.

He distinguishes:

True in every instance	true of all cases and always
Essential	belongs to what the subject is
Commensurately universal	belongs to the subject as such
Accidental	not essentially connected

Teaching Point

Scientific demonstration is concerned with what belongs essentially and universally, not with accidental conjunctions.

$$\boxed{\text{Demonstration proves essential attributes of their proper subjects.}}$$

13. True in Every Instance

An attribute is true in every instance when it is predicable of all instances of a subject and at all times.

Example

Animal is predicable of every human.

If this is a human, then this is an animal.

$$\text{Human} \Rightarrow \text{Animal}$$

Teaching Point

A scientific predicate cannot belong only occasionally or to only some cases.

$$\boxed{\text{Scientific demonstration requires stable predication.}}$$

14. Essential Attributes

An attribute is essential in two especially important ways.

First Sense

An attribute is essential when it belongs as an element in the subject's essential nature.

Line belongs to triangle.

Point belongs to line.

Second Sense

An attribute is essential when the subject is contained in the definition of the attribute.

Odd belongs to number.

Even belongs to number.

The definition of odd or even includes number.

Teaching Point

Essential attributes are not accidental add-ons. They are tied to what the subject is.

What belongs essentially belongs through the thing's nature.

15. Accidental Attributes

An accidental attribute is not connected with the subject through its essence.

Examples

The animal is white.

The person is musical.

These may be true, but they do not follow from what animal or human is.

Teaching Point

The accidental may be true, but it is not the proper object of scientific demonstration.

There is no demonstration of the accidental as accidental.

16. Commensurately Universal Attributes

An attribute belongs commensurately and universally when it belongs:

1. to every instance of the subject,
2. essentially,
3. as such,
4. to the first proper subject.

$$\boxed{\text{Commensurately universal} = \text{universal, essential, and proper}}$$

Triangle Example

Having angles equal to two right angles belongs to triangle as such.

It does not belong primarily to:

isosceles triangle

or:

scalene triangle

or:

equilateral triangle

It belongs first to:

triangle

Whiteboard

$$\text{isosceles} \rightarrow \text{triangle} \rightarrow \text{two right angles}$$

$$\boxed{\text{The proper subject is the first subject to which the attribute belongs as such.}}$$

17. The Error of the Wrong Universal

Aristotle warns that we may mistakenly think we have demonstrated something commensurately and universally when we have not.

Example

If we prove separately that:

equilateral triangles have angles equal to two right angles,

isosceles triangles have angles equal to two right angles,

scalene triangles have angles equal to two right angles,

we still have not grasped the attribute at the deepest scientific level unless we know that it belongs to triangle as such.

Teaching Point

Scientific knowledge seeks the correct level of universality.

$$\boxed{\text{Do not prove of a species what belongs first to the genus.}}$$

18. Necessary and Essential Premises

Because scientific knowledge concerns what cannot be otherwise, demonstration must proceed from necessary premises.

Essential attributes attach necessarily to their subjects. Accidental attributes do not.

$$\text{Essential} \Rightarrow \text{necessary}$$

$$\text{Accidental} \not\Rightarrow \text{necessary}$$

Example

A triangle has angles equal to two right angles.

This belongs necessarily.

This drawn triangle is blue.

This may be true, but not necessarily.

Teaching Point

Demonstrative premises must be necessary and essential.

$$\boxed{\text{Science demonstrates necessary connections through essential premises.}}$$

19. Demonstration Must Remain Within One Genus

Aristotle argues that demonstration cannot simply pass from one genus to another.

For example, geometry cannot prove geometrical truths by arithmetic, unless the sciences stand in a special subordinate relation.

$$\boxed{\text{Demonstration must use principles appropriate to its science.}}$$

The Three Elements of Demonstration

Every demonstrative science involves:

1. the subject genus
2. the axioms or basic premises
3. the attributes demonstrated

Teaching Point

Each science has its own domain, principles, and proper attributes.

$$\boxed{\text{Scientific knowledge is organized by genera.}}$$

20. Common Axioms and Proper Principles

Some truths are common to many sciences, but they are used only within the limits of a given science.

Example

If equals are subtracted from equals, equals remain.

This may be common, but the geometer uses it of magnitudes, while the arithmetician uses it of numbers.

Proper Principles

Each science also has proper principles peculiar to its subject matter.

$$\text{Geometry} \to \text{points, lines, figures}$$

$$\text{Arithmetic} \to \text{units, numbers}$$

Teaching Point

Common axioms do not erase the difference between sciences.

$$\boxed{\text{A science needs both common logical principles and proper subject principles.}}$$

21. Knowledge of the Fact and Knowledge of the Reasoned Fact

Aristotle distinguishes:

knowledge of the fact

from:

knowledge of the reasoned fact

The first tells us that something is so. The second tells us why it is so.

$$\boxed{\text{Scientific knowledge, strictly speaking, is knowledge why.}}$$

22. The Planets Example

Aristotle gives the example of the planets.

One might argue:

The planets do not twinkle.

Things that do not twinkle are near.

$\therefore$ The planets are near.

This proves the fact, but not the reasoned fact.

The true explanatory order is:

The planets are near.

Near heavenly bodies do not twinkle.

$\therefore$ The planets do not twinkle.

Teaching Point

The cause must be the middle term.

$$\boxed{\text{In demonstration, the middle term is not merely a connector; it is the cause.}}$$

23. The Moon Example

Another example concerns the moon.

One may infer:

$$\text{The moon waxes in this way.}$$

$$\therefore \text{The moon is spherical.}$$

But the deeper explanation is:

$$\text{The moon waxes in this way because it is spherical.}$$

Teaching Point

The fact may be more obvious to us, while the cause is prior by nature.

$$\boxed{\text{The reasoned fact reverses the order of mere observation.}}$$

24. Subordinate Sciences

Sometimes one science knows the fact while another knows the reason why.

Examples

$$\text{Optics} \to \text{subordinate to geometry}$$

$$\text{Harmonics} \to \text{subordinate to arithmetic}$$

$$\text{Observation} \to \text{subordinate to mathematical astronomy}$$

The empirical observer may know that something is so, while the mathematical science explains why.

Teaching Point

Some sciences are related as fact-gathering to cause-explaining.

$$\boxed{\text{Subordinate sciences receive their explanations from superior sciences.}}$$

25. The First Figure Is the Scientific Figure

Aristotle argues that the first figure is the most scientific.
Why?

1. It is the vehicle of mathematical demonstration.
2. It best gives the cause.
3. It enables knowledge of essence.
4. The other figures depend on it.

$$\boxed{\text{First figure} = \text{the figure of scientific explanation}}$$

Teaching Point

In the *Prior Analytics*, the first figure was the clearest form of syllogism. In the *Posterior Analytics*, it becomes the chief form of scientific demonstration.

$$\boxed{\text{The most scientific syllogism is the one that best displays the cause.}}$$

26. Ignorance as Error

Aristotle distinguishes ignorance as mere absence of knowledge from ignorance as positive error produced by inference.

$$\text{Ignorance as lack} = \text{not knowing}$$

$$\text{Ignorance as error} = \text{thinking falsely through inference}$$

Teaching Point

Bad reasoning can produce a positive state of ignorance.

$$\boxed{\text{Error is not merely absence; it can be false conviction.}}$$

27. Sense Perception and Induction

Aristotle says that if a sense is lacking, a corresponding kind of knowledge is also lacking.

Why?

Because induction depends on sense-perception, and universals are grasped through induction.

$$\text{sense perception} \to \text{particulars}$$

$$\text{particulars} \to \text{induction}$$

$$\text{induction} \to \text{universals}$$

$$\text{universals} \to \text{demonstration}$$

Teaching Point

There is no science without sense, but science is not mere sense.

$$\boxed{\text{Knowledge begins in perception and is completed in universal explanation.}}$$

28. The Regress of Premises Must Terminate

A major question in Book I is whether demonstration can involve an infinite regress of premises.

$$A \to B \to C \to D \to \cdots$$

If the chain never terminates, then there are no immediate premises.
If there are no immediate premises, then there are no first principles.
If there are no first principles, there is no demonstration.

$$\boxed{\text{Demonstration requires immediate premises; therefore regress must stop.}}$$

Teaching Point

Scientific knowledge depends on the termination of explanation in first principles.

$$\boxed{\text{An infinite chain of reasons would prevent completed knowledge.}}$$

29. The Superiority of Universal Demonstration

Aristotle argues that universal demonstration is superior to particular demonstration.

This does not mean that Aristotle believes in separated Platonic universals.

Rather, universal demonstration is superior because it gives the cause at the proper level.

Example

Particular demonstration:

Isosceles triangles have angles equal to two right angles.

Universal demonstration:

Triangles have angles equal to two right angles.

The universal demonstration is better because the attribute belongs to triangle as such, not to isosceles as such.

Teaching Point

The best demonstration proves the attribute of the first proper subject.

$\boxed{\text{The right universal gives the right cause.}}$

30. Universal Does Not Mean Over-General

Aristotle is not saying that the more general claim is always better.

The correct universal is the level at which the attribute belongs essentially.

Too low: isosceles

Proper: triangle

Too high: figure

Having angles equal to two right angles belongs to triangle as such, not to isosceles merely, and not to figure generally.

Teaching Point

Science seeks not the widest possible subject, but the proper explanatory subject.

$\boxed{\text{The scientific universal is exact, not merely broad.}}$

31. The Superiority of Affirmative Demonstration

Aristotle argues that affirmative demonstration is superior to negative demonstration.

Reasons include:

- affirmation is prior to negation,
- being is prior to not-being,
- negative demonstration depends on affirmative structure,
- affirmative demonstration is closer to the basic form of proof.

Teaching Point

Affirmative demonstration is more basic because it shows what belongs to what, not merely what does not belong.

$$\boxed{\text{Affirmation is prior to negation in scientific explanation.}}$$

32. Direct Demonstration and Reductio

Aristotle also ranks direct demonstration above *reductio ad impossibile*.

A direct demonstration proceeds from premises to conclusion.

$$P_1 + P_2 \Rightarrow C$$

A reductio assumes the contradictory and derives impossibility.

$$\text{not-}C \Rightarrow \bot$$

$$\therefore C$$

Teaching Point

Reductio may prove, but direct demonstration better displays the prior cause.

$$\boxed{\text{Direct explanation is superior to proof through impossibility.}}$$

33. Perception Is Not Demonstration

Scientific knowledge is not possible through perception alone.

Perception is of:

$$\text{this}$$

here

now

Science is of:

the universal

always

the cause

Example

Even if we perceived that a particular triangle has angles equal to two right angles, we would still seek demonstration.

Why?

Because perception gives the particular, while science grasps the commensurately universal.

$\boxed{\text{Seeing the fact is not the same as knowing the cause.}}$

34. The Eclipse Example

Aristotle says that even if we were on the moon and saw the earth blocking the sun's light, we would perceive the fact of the eclipse.

But we would not yet possess scientific knowledge unless we grasped the universal cause.

Perception: the earth is now blocking the sun.

Science: eclipses occur because light is obstructed in this way.

Teaching Point

Perception may start inquiry, but demonstration completes knowledge.

$\boxed{\text{Science requires universal causal explanation beyond perception.}}$

35. Different Sciences Have Different Basic Truths

Aristotle argues that not all sciences share the same basic truths.
Each science has principles appropriate to its own subject genus.

$$\text{Arithmetic} \to \text{number}$$

$$\text{Geometry} \to \text{magnitude}$$

$$\text{Medicine} \to \text{health and disease}$$

Teaching Point

There is no single stock of premises from which every scientific conclusion can be demonstrated.

$$\boxed{\text{Different sciences require different proper principles.}}$$

36. Opinion and Scientific Knowledge

Scientific knowledge concerns what is necessary and cannot be otherwise.
Opinion concerns what may be otherwise, even if the opinion is true.

$$\text{Knowledge} = \text{necessary}$$

$$\text{Opinion} = \text{can be otherwise}$$

Teaching Point

True opinion is not yet scientific knowledge.

$$\boxed{\text{To know scientifically is to grasp necessity, not merely truth.}}$$

37. Can One Know and Opine the Same Thing?

Aristotle says that knowledge and opinion may concern the same subject in a qualified sense, but not in the same way.

Example

One may know:

Human is essentially animal.

This is scientific knowledge if animal is grasped as part of the essence of human.
One may merely opine:

Human is animal.

if one does not grasp the essential and necessary connection.

Teaching Point

The difference lies not only in the object, but in the mode of apprehension.

$\boxed{\text{Knowledge grasps necessity; opinion treats the connection as contingent.}}$

38. Quick Wit

Book I ends with quick wit.
Quick wit is the ability to hit upon the middle term instantaneously.

$\boxed{\text{Quick wit} = \text{instant grasp of the middle term}}$

Examples

A person sees that the moon's bright side is always turned toward the sun and quickly grasps the cause:

The moon borrows its light from the sun.

A person sees someone talking with a wealthy man and quickly grasps:

He is borrowing money.

A person sees two people become friends and quickly grasps:

They share a common enemy.

Teaching Point

The whole book has taught us that the middle term is the key to demonstration. Quick wit is the ability to find that middle term immediately.

$\boxed{\text{To find the middle is to find the cause.}}$

39. The Architecture of Book I

The lecture can be summarized as the following movement:

$$\text{pre-existent knowledge} \to \text{scientific knowledge} \to \text{demonstration} \to \text{first principles} \to \text{essential attributes}$$

$$\to \text{appropriate sciences} \to \text{fact vs. reasoned fact} \to \text{first figure} \to \text{regress and principles}$$

$$\to \text{universal demonstration} \to \text{perception, opinion, quick wit}$$

Teaching Point

Book I turns formal logic into a theory of science.

$$\boxed{\text{The form of inference is not enough; science requires explanatory order.}}$$

40. Summary of Main Ideas

1. All teaching and learning by argument begins from pre-existent knowledge.
2. Scientific knowledge is knowledge of the cause as cause.
3. Demonstration is a syllogism productive of scientific knowledge.
4. Demonstrative premises must be true, primary, immediate, prior, better known, and explanatory.
5. Not all knowledge is demonstrative; first principles are indemonstrable.
6. Infinite regress and circular demonstration both fail to explain scientific knowledge.
7. Demonstration concerns necessary and essential attributes, not accidents as accidents.
8. The proper scientific universal is the first subject to which the attribute belongs as such.
9. Demonstration must proceed from principles appropriate to the subject genus.
10. Knowledge of the fact differs from knowledge of the reasoned fact.
11. In scientific demonstration, the middle term is the cause.
12. The first figure is the most scientific figure.

13. Universal demonstration is superior to particular demonstration when it gives the proper cause.
14. Affirmative demonstration is superior to negative demonstration, and direct demonstration is superior to reductio.
15. Perception begins inquiry, but scientific knowledge requires universal demonstration.
16. Opinion may be true, but scientific knowledge grasps necessity.
17. Quick wit is the immediate discovery of the middle term.

Final Framing

Posterior Analytics, Book I, turns logic into epistemology.

The *Prior Analytics* teaches us how conclusions follow. The *Posterior Analytics* teaches us when following becomes knowing.

A syllogism concludes; a demonstration explains.